Hackathon Prime: Turning PayPal's Weaknesses into Winning Strategies

PayPal AI Hackathon 2026 — Team Strategy Deck

For Beginners, By Strategists. This deck turns 3 PayPal complaints into 3 winning products. Simple language. Clear build plan. Prize-focused.

Team Level	Foundation / Beginners
Goal	Build Agentic Commerce that wins $12k Prize
Core Rule	PayPal API must be CORE, not a side feature

1. Executive Summary (For Beginners)

What is PayPal's main problem?

• PayPal is powerful but users complain about 3 things: money gets frozen suddenly, support is slow, and fees are confusing. These break trust.

What is Agentic Commerce?

• Imagine an AI assistant that doesn't just chat — it ACTS. It checks your account, predicts problems, talks to PayPal APIs, and fixes things for you automatically. That's an Agent.

Why will our approach win?

• We don't build random ideas. We fix PayPal's biggest pain points.

• Each idea has 2 levels: Tier 1 you can build in a weekend, Tier 2 wins the prize.

• We make PayPal API the hero — this is what judges MUST see to give us $12k.

■ **Winning Formula:** Real User Pain + AI Agent that Acts + PayPal API as CORE = Judges love it.

2. The Problem Landscape

We picked 3 problems that EVERY PayPal user has faced. Judges will instantly relate.

■ Problem 1: Unexpected Account Holds

Explain Like I'm 12: Think of your piggy bank being locked by your parents without warning. You did nothing wrong, but you can't use your own money. Small shops lose sales when PayPal freezes funds after a big order.

Why It Matters to PayPal (Business Impact):

- Merchants can't pay suppliers → business stops
- Trust drops, they move to Stripe/Razorpay
- PayPal spends more on manual reviews

■ Problem 2: Slow Customer Support

Explain Like I'm 12: Like waiting 3 days for teacher to check your homework while exam is tomorrow. When a buyer says 'I didn't get item', sellers wait days for human support to reply.

Why It Matters to PayPal (Business Impact):

- Disputes take days → money stuck
- Bad experience for buyers AND sellers
- High support cost for PayPal

■ Problem 3: Opaque or Confusing Fees

Explain Like I'm 12: Like a shop that says $10 but at counter charges $13. For international payments, users don't know currency conversion + cross-border + PayPal fee until money is gone.

Why It Matters to PayPal (Business Impact):

- Users feel cheated, avoid international sales
- Cart abandonment increases
- Competitors win with 'no hidden fees' messaging

3. Our Winning Solutions (The Core)

For each problem, we build TWO versions. Start with Tier 1, then level up to Tier 2 to win.

Solution 1: Proactive Health Dashboard — *Fixes: Account Holds*

TIER 1 — Basic (Weekend Build)
• Dashboard showing account status: Healthy / At Risk / Under Review
• Use PayPal Transaction API to show recent large/spike payments
• Simple traffic light UI: Green/Yellow/Red

TIER 2 — Advanced ($12k Winning)
• AI Agent scans transaction patterns daily (Gemini + anomaly detection)
• Predicts: 'Large $2k payment incoming — 78% chance of hold'
• Agent suggests action: Upload invoice / tracking BEFORE hold
• Auto-generates email draft with proof to PayPal
• PayPal API Core: Transaction Search + Webhooks + User Info

■ **Tech Highlight:** PayPal API: Transaction Search API, Webhooks, Orders API | AI: Gemini 1.5 Pro (pattern analysis), Python scikit-learn for anomaly | Sponsor: AG Grid for pro table

Solution 2: AI Dispute Copilot — *Fixes: Slow Support*

TIER 1 — Basic (Weekend Build)
• Chatbot that answers FAQs: 'How to respond to dispute?'
• Shows open disputes from PayPal Disputes API
• Template replies for common cases

TIER 2 — Advanced ($12k Winning)
• Autonomous Agent linked to Disputes API (24/7)
• When dispute filed: Agent instantly pulls evidence — transaction log, shipping proof, chat history
• Creates summary: 'Buyer claims not received, but tracking shows delivered'
• For simple cases: Auto-resolves in minutes via API (Provide Evidence endpoint)
• Human-in-loop for complex cases
• PayPal API Core: Disputes API + Orders + Tracking

■ **Tech Highlight:** PayPal API: Disputes API (List, Show, Provide Evidence), Orders API | AI: GPT-4o / Gemini for reasoning, LangChain Agent | Sponsor: AG Grid for dispute inbox

Solution 3: Transparent Fee Agent — *Fixes: Confusing Fees*

TIER 1 — Basic (Weekend Build)	TIER 2 — Advanced ($12k Winning)
• Simple fee calculator: Enter $100 to India → Shows PayPal fee $4.99 • Static table of fees by country • Clean Streamlit UI	• Agent analyzes payment BEFORE sending • Calculates all-in cost: PayPal fee + FX markup + cross-border + receiving fee • Shows in plain English: 'You send $100, friend gets $91.20. You pay $8.80 total.' • Suggests optimization: 'Send in EUR not USD to save $2.30' • What-if slider: Try different amounts/currencies instantly • PayPal API Core: Payouts + Exchange Rate (via PayPal API), Orders API for checkout simulation

■ **Tech Highlight:** PayPal API: Payouts API, Orders API, Exchange Rate estimate | AI: Gemini for plain-language explanation, function calling | Sponsor: AG Grid for comparison table

4. Why We Will Win — Judging Criteria Map

Judges have 4 criteria. We hit all 4 AND we obey the 'PayPal must be CORE' rule.

Criteria	How We Hit It	Our Edge
Technological Implementation (30%)	Not just chatbot. Real Agent that calls PayPal APIs, acts autonomously, uses webhooks.	Live API calls > Mock data
Impact (30%)	We solve money-freeze, dispute delay, fee confusion — PayPal loses users over these.	Business ROI clear
Innovation (20%)	Proactive not reactive: Predict hold BEFORE it happens, resolve dispute in minutes, show fees BEFORE send.	Agentic Commerce = Future
Design (20%)	Traffic light dashboard, plain English fees, dispute inbox with AG Grid — simple for beginners, pro for judges.	Beginner-friendly UX

■■ **PayPal CORE Rule — How We Pass:** PayPal API is not an add-on. Without PayPal Transaction/Disputes/Payouts API, our product DOES NOT WORK. Every feature starts with a PayPal API call. This is what judges check first. We put PayPal logo + API call log visible on UI.

5. Tech Stack For Beginners

Use only tools you already know. No heavy DevOps.

Layer	Tool (Beginner Friendly)	Why This?
Frontend	Streamlit (Python) OR React + Tailwind	Streamlit = 1 file, fast. React = looks pro for finals.
Backend	Python FastAPI OR Node.js	Handles PayPal webhooks + AI calls
PayPal API	Transaction Search, Disputes, Payouts, Orders, Webhooks	CORE — must be in every flow
AI Tool	Gemini 1.5 Pro (free tier) OR OpenAI GPT-4o-mini	Gemini great at analysis, OpenAI good agent tool-calling
Agent Framework	LangChain OR Direct Function Calling	Lets AI decide which PayPal API to call
Sponsor Tool	AG Grid (for tables)	Sponsor prize + makes dashboard look enterprise-grade
Hosting	Streamlit Cloud / Vercel / Render	Free, 5-min deploy for demo

Beginner Starter Command:

pip install streamlit paypalrestsdk google-generativeai langchain ag-grid → build Tier 1 in Day 1

6. Next Steps / Roadmap — 3 Week Plan

Timeline	Tasks	Outcome
Week 1: Learn & Build Basic	• Get PayPal Sandbox keys • Build Tier 1 for 1 problem (pick Dispute Copilot — easiest) • Show data in Streamlit + AG Grid	Working demo you can show
Week 2: Add AI Agent	• Connect Gemini / GPT-4o • Agent calls PayPal API automatically • Add Tier 2: prediction / auto-resolve / fee explanation • Record 30-sec demo video	Agentic version ready
Week 3: Polish & Win	• Make PayPal CORE obvious: API log panel on UI • Design: traffic lights, plain English, mobile responsive • Write README mapping to judging criteria • Submit + practice pitch: 'We turn PayPal weakness into strength'	$12k submission ready!

Pro Tip for Beginners: Don't build all 3 solutions. Pick ONE (we recommend #2 Dispute Copilot) and build Tier 1 + Tier 2 perfectly. Judges prefer 1 polished product over 3 half-baked.

Let's Build. ■

You are not just fixing bugs. You are showing PayPal the future of Agentic Commerce.

Remember:

• Start simple, then add AI magic
• Every screen must show a PayPal API call
• Talk like you're explaining to a 12-year-old — judges love clarity
• Your weakness as beginners is your strength: you build what real users need, not what engineers think is cool.

Now open Streamlit, get Sandbox keys, and ship Tier 1 today. You've got this.